# Rotation-Equivariant Variational Autoencoders for Representation Learning and Downstream Classification in Histopathology

**Maximiliano Garavito Chtefan** MAXIMILIANO2162094@CORREO.UIS.EDU.CO
**David Romo-Bucheli** DEROMOB@UIS.EDU.CO
*Escuela de Ingeniería de Sistemas e Informática*
*Universidad Industrial de Santander, Bucaramanga, Colombia*

**Editors:** Under Review for MIDL 2027

## Abstract

At high magnification, cellular patterns in histopathology have no dominant orientation in the image, making rotation a relevant local symmetry for representation learning. This work investigates whether incorporating this rotation symmetry into unsupervised representation learning improves reconstruction and performance in downstream classification tasks. We first train a conventional denoising variational autoencoder (VAE) and a continuous-SO(2)-equivariant denoising VAE trained on the same unlabeled patches from ovarian cancer images. With both autoencoders models' weights frozen, reconstruction was evaluated on patches from held-out whole-slide images (WSIs). Afterwards, we evaluated the representation capability of both autoencoders in two downstream tasks: three-class tissue recognition at patch level, and five-class WSI diagnosis via multiple-instance learning. Reconstruction mean absolute error for the test patches was 0.0629 versus 0.0642 (conventional versus equivariant), but the difference was not statistically significant. For three-class tissue recognition with a small budget of 500 training labels per class, the equivariant VAE outperforms the conventional VAE with a macro-F1 of 0.511 versus 0.575. For other budgets tested (250, 1000, 2500 and 5671) no statistical difference was found, although the equivariant VAE perfomed better in all the budgets except for the largest one. Finally, for WSI classification on 152 test slides, macro-F1 was $XXXX$ versus $YYYY$ (conventional versus equivariant), with no statistically significant difference.

**Keywords:** histopathology, variational autoencoder, rotation equivariance, SO(2), whole-slide image classification, representation learning

## 1. Introduction

Digital histopathology allows for digitization of glass slides into whole-slide images whose dimensions commonly reach gigapixel scale. These images contain many local tissue regions, but their diagnostic labels and detailed annotations require specialist review. Representation learning via unlabeled patches allows machine learning models to extract useful representations for using in a downstream task. (Litjens et al., 2017; Song et al., 2023; Tellez et al., 2021).

While orientation can be relevant to tissue architecture at lower magnification, cellular patterns at high magnification have no dominant orientation in the image. At this scale, a rotation changes the viewing orientation of a patch while preserving its tissue identity and diagnostic information. This rotational symmetry motivates incorporating rotation

© 2027 CC-BY 4.0, M.G. Chtefan & D. Romo-Bucheli.

into the model's architectural constraints. Ordinary convolutional neural networks (CNNs) use convolutions that are equivariant to translations: translating an input translates its computed feature map (LeCun et al., 1998; Cohen and Welling, 2016). Conventional CNNs usually learn rotational responses from rotated examples or separate filters, if the image domain is equivariant, they are expected to learn the equivariance as a byproduct of the training process. In contrast, the SO(2) VAE uses steerable convolutions with constrained filters to incorporate rotation symmetry as a prior, introducing a rotational inductive bias into supervised learning tasks (Cohen and Welling, 2016, 2017; Weiler and Cesa, 2019).

Rotation-equivariant classifiers have previously been studied for histopathology patch prediction (Veeling et al., 2018). Using equivariance to 90° rotations and reflections, Veeling et al. reported 89.8% accuracy and 96.3% ROC-AUC on PatchCamelyon, compared with 87.6% and 95.5%, respectively, for a conventional DenseNet. VAEs incorporating symmetry have also modeled object pose and learned orientation-invariant cell representations. On uniformly rotated and translated MNIST images, TARGET-VAE achieved 63.4% clustering accuracy with its $P_{16}$ variant, compared with 12.8% for a conventional VAE (Nasiri and Bepler, 2022). On the synthetic PCST cell benchmark, O2-VAE reduced the rate at which an image and its rotated copy were assigned to different clusters from 58% with a conventional VAE to 7.4% (Burgess et al., 2024). Lafarge et al. (2020) incorporated rotation-equivariant convolutions into VAEs for orientation-disentangled representation learning in histopathology. However, to our knowledge, no prior study has jointly evaluated spatial representations from continuous-SO(2) denoising VAEs for reconstruction, tissue recognition, and WSI diagnosis in ovarian histopathology.

For this study, we use patches from the UBC Ovarian Cancer Subtype Classification and Outlier Detection (UBC-OCEAN) training resource (University of British Columbia, 2023). A conventional residual denoising VAE and a denoising VAE with steerable kernels constrained for the continuous planar rotation group SO(2) are trained without supervision (Kingma and Welling, 2014; Im et al., 2017). Both yield spatial feature maps of size $16 \times 32 \times 32$.

This study examines two questions:

1. Does the implemented rotation-equivariant VAE yield lower reconstruction error on unseen histopathology patches than the conventional VAE?

2. Does the rotation-equivariant VAE representation yield higher performance (as measured by macro-F1) in downstream classification tasks: (a) for patch-level three-class tissue recognition from individual latent maps across label budgets, and (b) for WSI-level five-class diagnosis from bags of latent maps?

## 2. Related Work

### 2.1. Unsupervised Representation Learning for Histopathology

Unsupervised learning uses unlabelled histopathology patches to learn representations that can subsequently support supervised prediction. Autoencoders learn these representations through reconstruction, while downstream tasks assess whether they retain information relevant to tissue recognition and slide-level diagnosis. Tellez *et al.* used neural image

compression to learn patch representations without supervision before predicting gigapixel image labels. Their comparison included reconstruction, contrastive training and adversarial feature learning (Tellez et al., 2021).

Lafarge *et al.* extended the VAE framework with rotation-equivariant convolutions to separate oriented and isotropic latent components in histopathology images. They evaluated representations of individual cells and aggregated cellular representations for downstream assessment of tissue regions (Lafarge et al., 2020).

These studies motivate learning features without supervision and assessing their relevance through downstream tasks. Here, the VAEs are trained on unlabeled histopathology patches, evaluated in reconstruction, and frozen before their representations are used for patch-level tissue recognition and WSI-level diagnosis.

### 2.2. Equivariance for Deep Learning

Equivariance specifies how a learned representation changes when its input is transformed. For an input transformation $T_g$, an equivariant mapping satisfies $f(T_g x) = T'_g f(x)$, where $T'_g$ is the corresponding transformation of the output. Invariance is the special case in which the output is unchanged (Cohen and Welling, 2016). Figure 1 illustrates this relation for rotations: the output action rotates spatial locations and, for vector-valued fields, vector components.

By sharing filters across transformations, equivariant networks reuse learned patterns in different poses, reducing the need to learn separate detectors for transformed copies. This inductive bias can improve data efficiency when the chosen transformations preserve the structure relevant to the task (Cohen and Welling, 2016). Elesedy and Zaidi proved a generalisation benefit for linear invariant and equivariant models when the target distribution respects the prescribed compact-group symmetry (Elesedy and Zaidi, 2021).

Ordinary CNNs implement translation-equivariant convolutions through spatial weight sharing. Group convolutions extend this construction to additional transformations, including discrete rotations and reflections (Cohen and Welling, 2016). Steerable CNNs support continuous rotations by constraining convolution kernels according to the transformation laws of their input and output fields. The kernels are parameterised as learned linear combinations of admissible basis functions; normalisation, nonlinearities and pooling must also respect the field transformation laws (Cohen and Welling, 2017; Weiler and Cesa, 2019). Previous histopathology work applied rotation-equivariant networks to supervised patch classification (Veeling et al., 2018; Graham et al., 2020).

VAEs incorporating these principles provide precedents in other imaging domains. Nasiri and Bepler proposed TARGET-VAE, combining a translation- and rotation-equivariant encoder with a spatially equivariant generator to disentangle pose from an invariant semantic representation. They evaluated the method on MNIST digits, synthetic dSprites shapes, galaxy images and cryo-electron microscopy particle images (Nasiri and Bepler, 2022).

Burgess *et al.* introduced O2-VAE for cell and organelle profiling using two-dimensional segmentation masks and greyscale images derived from fluorescence microscopy, including confocal data, alongside a synthetic cell-shape and texture benchmark. O(2)-equivariant layers and pooling produce orientation-invariant latent vectors, while the reconstruction loss accounts for orientation differences (Burgess et al., 2024).

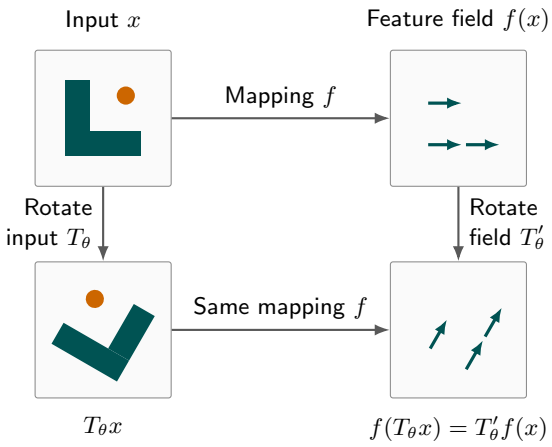



Figure 1: Rotation equivariance as a transformation law. The two paths from $x$ to the lower-right field agree for an equivariant mapping $f$. The input rotation $T_\theta$ acts on the image; the corresponding output action $T'_\theta$ also rotates vector components when the output contains vector fields. Shapes and arrows are schematic rather than measured VAE features.

### 2.3. Multi-Instance Learning

Whole-slide images can contain billions of pixels. Their size makes processing at native resolution in a single forward pass computationally impractical. A common approach therefore divides each WSI into patches, learns a local representation, and aggregates patch features for slide-level prediction. Attention-based multiple instance learning (MIL) provides a related supervised formulation in which a WSI is treated as a bag of instances and only the bag label is required (Ilse et al., 2018). CLAM combines attention pooling with instance-level clustering (Lu et al., 2021). DSMIL combines contrastive patch representation learning with dual-stream aggregation (Li et al., 2021), whereas TransMIL models relations among patch instances during aggregation (Shao et al., 2021). Here, MIL tests the frozen patch representations under the same WSI classification protocol. Figure 2 illustrates the slide-to-bag formulation.

## 3. Materials and Methods

### 3.1. UBC-OCEAN dataset

The UBC-OCEAN ovarian-cancer training resource contains 538 slide images: 513 conventional whole-slide images (WSIs) and 25 tissue microarray (TMA) images (University of British Columbia, 2023). A WSI shows a large tissue section; a TMA shows multiple small tissue cores in one image. The resource describes WSIs as large, approximately 20× scans and TMAs as smaller, approximately $4000 \times 4000$-pixel, 40× images. This scale and layout difference motivated using only the 513 non-TMA WSIs in the present experiment. Figure 3 shows a representative image of each type and their original pixel dimensions. For an

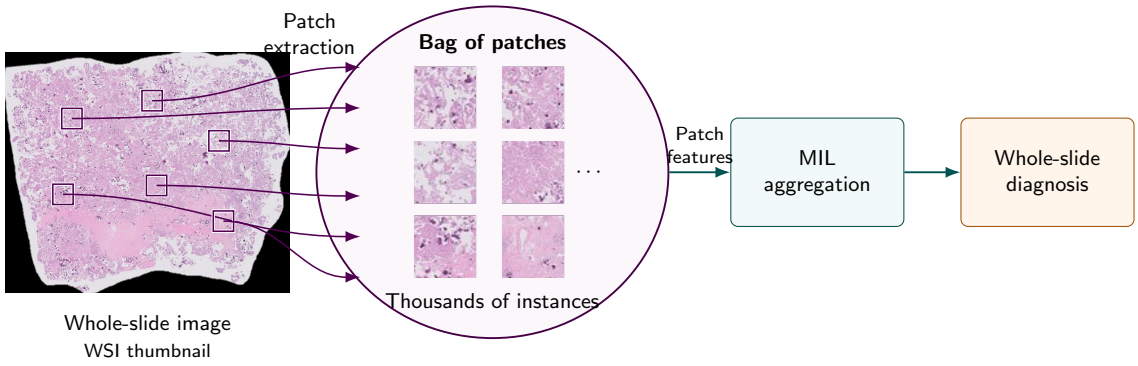



Figure 2: Multiple-instance learning for WSI diagnosis. Many patches extracted from one slide form a bag. Their encoded features are aggregated to predict one whole-slide diagnosis. The six thumbnail regions illustrate the slide-to-bag construction; the ellipsis represents the many remaining instances. In this study, patch features are obtained from frozen VAE posterior means, and supervision is provided by the WSI diagnosis label.

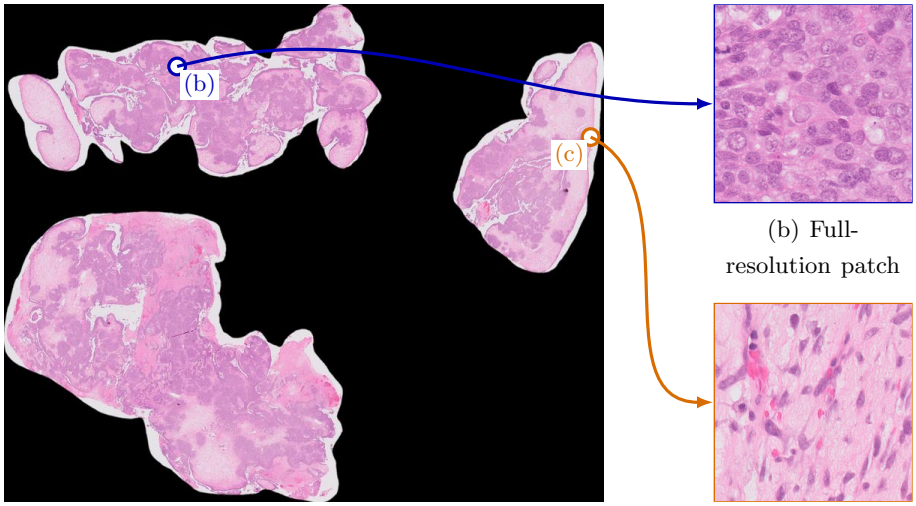
image of width $W$ and height $H$, a non-overlapping 256-pixel grid contains approximately $\lfloor W/256 \rfloor \lfloor H/256 \rfloor$ full patches before foreground filtering. At the dimensions quoted by the resource, a $4000 \times 4000$ image yields about 225 full patches, whereas a $100000 \times 50000$ image yields about 76,000; the latter is an illustrative large-WSI scale, not a count for an individual study slide. The resource also provides 152 supplemental color masks paired with a subset of those WSIs. These masks mark selected tumor, stroma and necrosis regions; they do not annotate every tissue pixel. The other 361 WSIs have no supplemental mask. The diagnostic label of each image is one of clear cell carcinoma (CC), endometrioid carcinoma (EC), high-grade serous carcinoma (HGSC), low-grade serous carcinoma (LGSC), or mucinous carcinoma (MC).

Each WSI was divided into non-overlapping RGB $256 \times 256$-pixel patches. A full WSI has one diagnosis label, so thousands of patches extracted from that slide share the same slide-level label. Local tissue labels, in contrast, come only from eligible regions of the supplemental masks. Splits were made by WSI: no patches from one slide crossed train, validation or test boundaries.

Figure 3 shows two full-resolution patches beside their source WSI thumbnail, a partial supplemental mask on a second WSI, and a TMA. The thumbnail makes the slide-to-patch scale change visible; model inference used crops from the full-resolution WSI. The mask marks selected tissue regions rather than the full image.

### 3.2. Patch Selection and Tissue Annotations

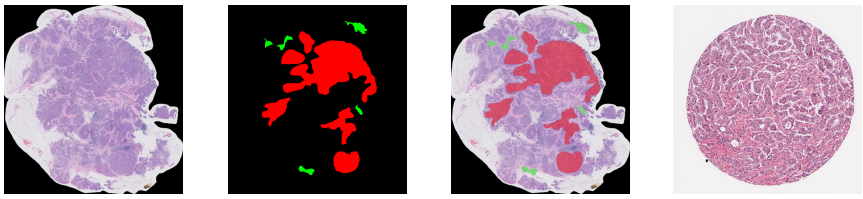
Foreground selection for reconstruction and WSI diagnosis was derived from the image rather than from human masks. Each WSI thumbnail was converted to HSV, and Otsu's threshold was applied to the saturation channel (Otsu, 1979). A full-resolution grid cell was retained when more than 60% of its projected thumbnail area belonged to foreground. The




(a) WSI thumbnail

(b) Full-resolution patch

(c) Full-resolution patch



(d) WSI thumbnail (e) Partial mask (f) Aligned overlay (g) TMA

Figure 3: Images and annotations in UBC-OCEAN (University of British Columbia, 2023; sohier, 2023). Panels (a)–(c) show a WSI thumbnail and two actual $256 \times 256$ crops from that WSI; arrows mark their source regions. Panels (d)–(f) show a different, paired WSI thumbnail, supplemental tissue mask and overlay. Red marks tumor, green stroma and blue necrosis when present; black mask pixels are unannotated. Panel (g) shows a TMA. The WSI in (d) measures $40063 \times 42207$ pixels, compared with $2964 \times 2964$ for the TMA. The panels do not share a physical scale; only WSIs enter the experiment.

resulting coordinates were used for VAE development, reconstruction and WSI diagnosis without consulting the supplemental tissue masks.

Supplemental masks were used only to construct the tissue-recognition population. Red, green and blue mask regions were interpreted as tumor, stroma and necrosis. The masks are not exhaustive; an unpainted region cannot be treated as a negative class. A patch was therefore eligible for tissue classification only when at least 10% of its area was annotated and at least 95% of that annotated area belonged to one tissue class. The resulting task classifies high-purity annotated regions and does not constitute dense pixel-level segmentation of the complete WSI.

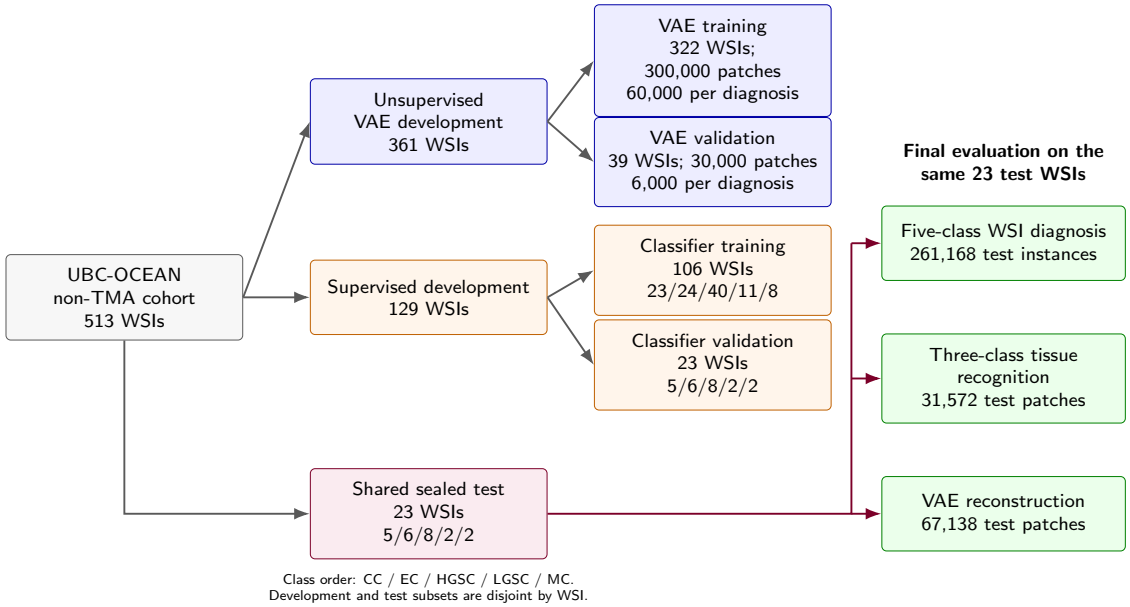



Figure 4: Data populations and WSI-level partitions. The source cohort is split into unsupervised VAE development, supervised classifier development, and a separate shared sealed test. The same 23 test WSIs support all three final evaluations, with task-specific patch populations. All assignments are fixed at WSI level and shared by both representations. Numbers inside split boxes follow the order CC/EC/HGSC/LGSC/MC.

### 3.3. Data Populations and Partitions

The 513 eligible WSIs were partitioned at WSI level into three disjoint subsets: 361 for unsupervised VAE development, 129 for supervised classifier development, and 23 for a shared sealed test (Figure 4). VAE development used 322 training and 39 validation WSIs, contributing 300,000 and 30,000 unlabeled patches. Patch sampling supplied 60,000 training and 6,000 validation patches from each WSI diagnosis; diagnosis was not a VAE training target. WSI counts were not balanced by diagnosis (Table 1). Supervised development used 106 training and 23 validation WSIs. Together with the shared test, these formed the 152-WSI cohort with supplemental masks. These assignments were fixed before VAE or classifier outputs on that cohort were inspected. They were shared by both downstream tasks and used diagnosis and annotated-tissue availability, but not image pixels, VAE representations or model predictions. The same 23 test WSIs were also used for reconstruction evaluation.

For WSI diagnosis, each slide was one labeled MIL bag containing all its foreground patches selected by Otsu thresholding. Each complete WSI contributed one diagnosis label shared by all patches in its bag. The patch-level tissue classifier instead used high-purity annotated tumor, stroma and necrosis patches. Its training subsets were balanced at the five nested label budgets introduced above, whereas validation and test used fixed, unbalanced patch sets. Table 1 separates the inputs and class counts of these three stages.

Table 1: Inputs by task and split. Diagnosis order is CC/EC/HGSC/LGSC/MC; tissue order is tumor/stroma/necrosis. VAE patch diagnosis is inherited from the source WSI, not a local tissue label. MIL class counts refer to WSIs; tissue class counts refer to patches.

| Task | Split | WSIs | Patches | Class composition |
|---|---|---|---|---|
| VAE | Train | 322 | 300,000 | 54/74/144/24/26 WSIs; 60,000 patches per diagnosis. |
| | Validation | 39 | 30,000 | 7/9/17/3/3 WSIs; 6,000 patches per diagnosis. |
| Reconstruction | Test | 23 | 67,138 | Foreground patches; uncertainty aggregated by WSI. |
| Tissue | Train | 106 | 750–17,013 | 250–5,671 selected patches per tissue class. |
| | Validation | 23 | 31,339 | 22,255/7,809/1,275 patches. |
| | Test | 23 | 31,572 | 21,796/8,500/1,276 patches. |
| WSI MIL | Train | 106 | 1,224,875 | 23/24/40/11/8 WSI labels; complete foreground bags. |
| | Validation | 23 | 264,178 | 5/6/8/2/2 WSI labels; complete foreground bags. |
| | Test | 23 | 261,168 | 5/6/8/2/2 WSI labels; complete foreground bags. |

### 3.4. Experimental Design

The same frozen VAE pair was evaluated in three tasks. First, reconstruction measured whether each encoder–decoder recovered clean, unseen patches. Second, a patch classifier used frozen latent maps to predict local tissue type. It was trained at five nested label budgets, with the same validation and test sets at every budget. Third, a multiple-instance learning (MIL) classifier aggregated patch embeddings to predict one diagnosis per WSI.

The VAEs were trained on unlabeled patches and frozen before classification. Reconstruction used each encoder and decoder; classification used only the deterministic posterior mean. Both VAEs received the same ordered patches, normalization, corruption policy, loss, update count and validation schedule. Downstream pairs shared WSI assignments, patch coordinates, classifier initialization, epoch order and model-selection rule. The architectures differ in parameter count and computational cost (Table 2).

### 3.5. Variational Autoencoders and Shared Architecture

An autoencoder learns to turn an image into a smaller representation and then reconstruct the image from it. Its *encoder* extracts and compresses visual features, and its *decoder* uses the compressed embedding to reproduce the input. In a VAE, the encoder specifies a distribution over embeddings rather than one fixed code: here a diagonal Gaussian with mean $\boldsymbol{\mu}$

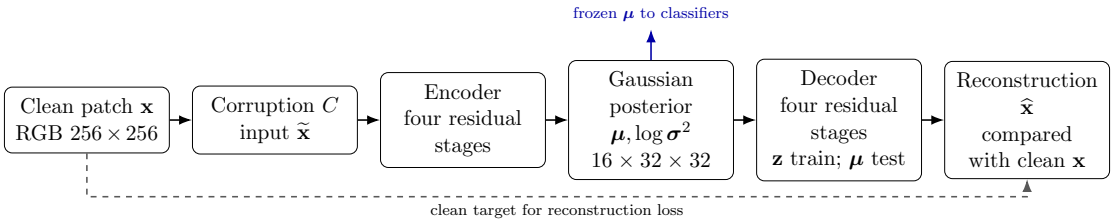



Figure 5: Shared denoising VAE workflow. Both models have the same input, posterior shape and encoder–decoder resolution schedule. They differ in the type of convolution and the accompanying field normalization and gates. Training reconstructs the clean patch from a possibly corrupted input; classification uses only the frozen posterior mean. The dashed rectangular path supplies the clean input as the reconstruction target.

and log-variance $\log \boldsymbol{\sigma}^2$. During training, a latent sample $\mathbf{z}$ is drawn by reparameterization (Kingma and Welling, 2014). The distribution is regularized toward a unit Gaussian while the decoder learns image reconstruction. After training, the posterior mean is a deterministic feature map for the classifiers. For reconstruction evaluation, the decoder reconstructs the clean patch from that mean. Figure 5 shows these paths.

Both models used a four-stage residual encoder informed by ResNet-18 (He et al., 2016): a stem and two residual blocks at each stage, with $2\times$ fixed anti-aliased downsampling between stages. Their decoder used two residual blocks per stage and reversed the three resolution changes with bilinear upsampling followed by convolution. The spatial sequence was $256 \to 128 \to 64 \to 32$ pixels per side in the encoder and reversed in the decoder. Separate posterior heads produced

$$\boldsymbol{\mu}, \log \boldsymbol{\sigma}^2 \in \mathbb{R}^{16\times 32\times 32}. \tag{1}$$

The decoder returned a raw three-channel image from a zero-initialized output convolution, without a final tanh. The 16-channel spatial embedding preserves coarse locations within the patch; its $32 \times 32$ grid is smaller than the original $256 \times 256$ input. Table 2 compares the two implementations.

### 3.6. Continuous Rotation Equivariance

A *group* is a collection of transformations that can be composed and inverted. The planar rotation group SO(2) contains every angle from 0 to 360 degrees, with angle addition as its composition rule. A map $f$ is *equivariant* when transforming its input also transforms its output in a predictable way:

$$f(T_g \mathbf{x}) = \rho(g) f(\mathbf{x}), \qquad g \in \mathrm{SO}(2). \tag{2}$$

Here $T_g$ rotates an image and $\rho(g)$ specifies how the output feature field must rotate. This is distinct from *invariance*, in which the output stays identical after rotation. For a scalar image feature, spatial locations rotate but the value at a corresponding location does not; for a vector feature, its two components rotate as well.

The conventional VAE used ordinary convolutions, GroupNorm and learned scalar gates. Its hidden channel counts were 32, 48, 64 and 96. The experimental SO(2) VAE replaced the convolutional layers with steerable convolutions (Cohen and Welling, 2017; Weiler and Cesa, 2019). A steerable kernel is a filter whose rotated version follows a prescribed rule: rather than learning an unrelated filter for every orientation, the network learns coefficients of radial and angular basis functions that obey the input/output field transformation laws. This construction constrains the convolutional operators; it does not by itself establish that the complete trained encoder obeys Equation 2 exactly under sampled image rotations.

Its hidden features contained $F_0$ scalar fields and $F_1$ vector fields. An $F_0$ copy stores one value per pixel and changes only its spatial location under a rotation. An $F_1$ copy stores an ordered pair of values per pixel; that pair rotates by the same angle as the image coordinate system. Thus, a layout with $n_0$ scalar copies and $n_1$ vector copies has $n_0 + 2n_1$ numerical channels, not $n_0 + n_1$. The four stages used $(n_0, n_1) = (16, 16), (24, 24), (32, 32), (48, 48)$. Field-aware normalization and gates based on the $F_1$ vector magnitude were used so these operations respect the field types. The precise nonlinearities and normalizations are given below. The posterior heads returned 16 $F_0$ fields and no $F_1$ fields, matching the conventional $16 \times 32 \times 32$ posterior.

#### 3.6.1. Kernel Support and Learned Parameterization

Comparing conventional and rotation-equivariant CNNs is not straightforward: matching kernel support, learned parameter count and computational cost can require different architectural choices. The conventional VAE uses $5 \times 5$ residual kernels, whereas the SO(2) VAE uses $7 \times 7$ steerable kernels. This difference follows from the spatial sampling requirements of the selected steerable basis.

The basis combines angular harmonics with Gaussian radial profiles, a common construction in steerable CNNs (Weiler and Cesa, 2019). The radial envelopes are centered at approximately 1.00, 1.90 and 2.75 pixels, each with a standard deviation of 0.30 pixels. Its $F_1$-to-$F_1$ terms include a second angular harmonic, corresponding to two cycles around the filter center. A centered $5 \times 5$ grid extends only two pixels along each coordinate axis; only its four corners lie beyond radius 2.5 pixels. Consequently, the outer radial envelope is sampled unevenly across angles. The $7 \times 7$ grid extends three pixels along each axis and provides additional samples near this envelope, improving its angular coverage and reducing the risk of discretization aliasing.

While the larger grid might appear to give the steerable layers an advantage, their filters remain constrained to a fixed steerable basis. They learn coefficients of this basis rather than an independent weight at every kernel location. For example, the steerable convolution in the first same-resolution residual stage uses only 30.0% as many learned parameters as its conventional counterpart: 7,680 basis coefficients versus 25,600 free weights. Across the complete VAEs, the SO(2) model has 29.8% as many learned parameters, although its estimated convolutional work is higher (Table 2).

Larger steerable than conventional kernels have also been used in earlier comparisons. Weiler and Cesa used larger steerable kernels to mitigate discretization artifacts while adjusting feature widths to preserve the total parameter count (Weiler and Cesa, 2019). Worrall *et al.* compared $5 \times 5$ harmonic filters with $3 \times 3$ conventional filters and reported their

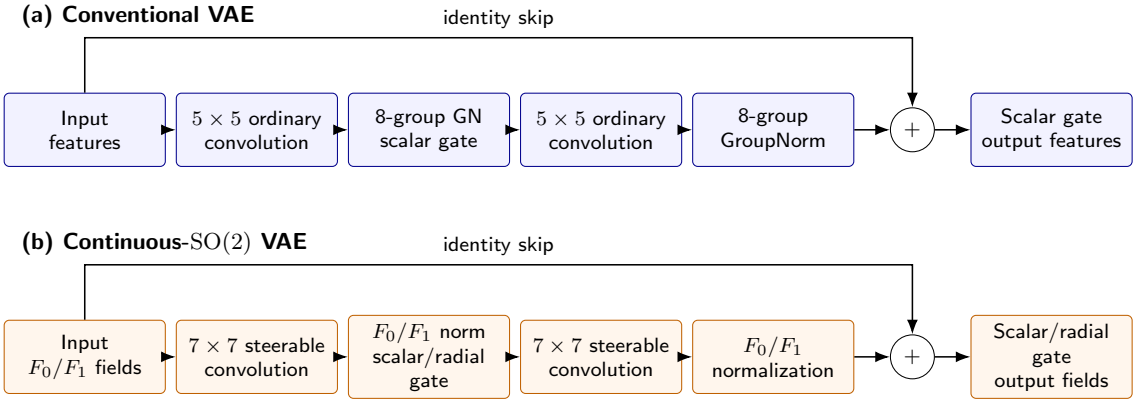



Figure 6: Same-resolution residual blocks used by the conventional and continuous-SO(2) VAEs. Each main branch contains two learned convolutions; normalization and a gate follow the first convolution, while the second is followed by normalization. The identity skip is added before the output gate. At encoder stage transitions, both branches are downsampled separately with a fixed $5 \times 5$ binomial filter: on the main branch after the first gate and on the skip branch before its learned projection and normalization. At decoder transitions, both branches are bilinearly upsampled before their first learned convolution; the skip branch again uses a projection and normalization. The $F_0/F_1$ field types and gate equations are defined in the surrounding text. The $7 \times 7$ steerable support provides fuller sampling of the selected radial and angular basis; Section 3.6.1 gives the rationale and learned-parameter comparison.

different parameter counts (Worrall et al., 2017). Figure 6 shows the two residual building blocks before Figure 7 assembles them into the full encoder and decoder schedules.

### 3.7. Field Gates and Normalization

The conventional VAE applies a learned scalar gate to each hidden channel, including after residual addition. This gate is a parameterized variant of the sigmoid-weighted linear unit (SiLU), $u\,\sigma(u)$ (Elfwing et al., 2018), with a learned slope and offset inside the sigmoid. For a channel value $u_c$,

$$A_0(u_c) = u_c\,\sigma(a_c u_c + b_c), \qquad \sigma(t) = \frac{1}{1+e^{-t}}, \tag{3}$$

where $a_c$ and $b_c$ are learned per-channel parameters, initialized to $a_c = 1$ and $b_c = 0$, so the initial activation is SiLU. The $F_0$ fields in the SO(2) VAE use the same rule with one parameter pair per scalar field. For an $F_1$ field $\mathbf{v}_j = (v_{j,1}, v_{j,2})$, the equivariant VAE gates both components by the same factor computed from their rotation-invariant magnitude:

$$\begin{aligned} r_j &= \sqrt{v_{j,1}^2 + v_{j,2}^2 + 10^{-4}}, \\ A_1(\mathbf{v}_j) &= \mathbf{v}_j\,\sigma(\alpha_j r_j + \beta_j). \end{aligned} \tag{4}$$

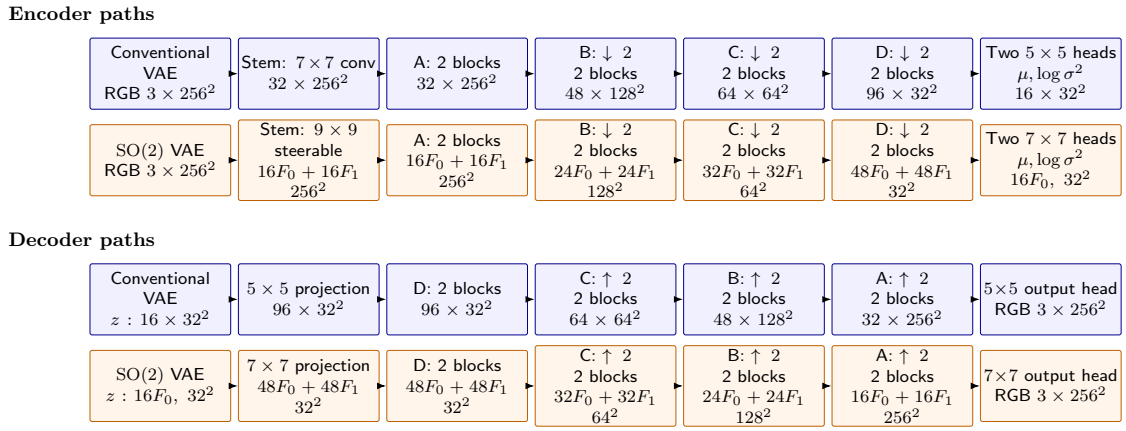



Figure 7: Layer-level architecture of the two frozen VAEs. The residual building blocks are detailed in Figure 6. Each stage A–D contains two residual blocks; stage-changing blocks additionally resample both branches and project the skip branch. A conventional hidden width is a channel count; in the SO(2) model, $nF_0 + nF_1$ means $n$ scalar fields and $n$ two-component vector fields. Posterior mean and log-variance use separate heads; training samples $\mathbf{z}$, whereas downstream classifiers use $\boldsymbol{\mu}$. The conventional stem uses a $7 \times 7$ kernel and its other convolutions use $5 \times 5$ kernels; the SO(2) stem uses a $9 \times 9$ steerable kernel and its other convolutions use $7 \times 7$ kernels. See Section 3.6.1 for the kernel-support rationale and learned-parameter comparison.

The learned $\alpha_j, \beta_j$ are shared by the two components. A planar rotation changes the vector direction but not $r_j$, so this gate rotates its output together with the input vector.

Before these gates, the conventional network uses affine GroupNorm with eight groups. For each sample and channel $c$, it computes the mean and variance over the channels in $c$'s group and all spatial positions:

$$N_{\mathrm{GN}}(u_c) = \gamma_c \frac{u_c - \mu_{G(c)}}{\sqrt{s^2_{G(c)} + 10^{-5}}} + \delta_c. \tag{5}$$

The equivariant network applies this same eight-group formula only to its $F_0$ scalar fields. Its $F_1$ vectors instead use four groups and a shared root-mean-square denominator for both components:

$$N_1(\mathbf{v}_j) = \gamma_j \frac{\mathbf{v}_j}{\sqrt{\underset{k \in G(j),\, d \in \{1,2\},\, p}{\mathrm{mean}} v^2_{k,d,p} + 10^{-5}}}. \tag{6}$$

Here $p$ indexes spatial positions within one sample. The vector rule has no mean subtraction or additive bias, because either would introduce a preferred direction. Its denominator is unchanged by rotations in feature space. The normalization statistics and gates are evaluated in FP32 in the SO(2) implementation.

Table 2: Architecture comparison.

| Property | Conventional VAE | Continuous-SO(2) VAE |
|---|---|---|
| Hidden representation | Ordinary channels: 32, 48, 64, 96 | $(F_0, F_1)$ copies: (16,16), (24,24), (32,32), (48,48); 48, 72, 96, 144 numerical channels |
| Learned convolutions | 43 ordinary convolutions | 43 steerable convolutions |
| Normalization/ nonlinearity | Eight-group GroupNorm; learned scalar gate | Eight-group $F_0$ GroupNorm; four-group $F_1$ RMS normalization; scalar and radial gates |
| Posterior | $\boldsymbol{\mu}, \log \boldsymbol{\sigma}^2 \in \mathbb{R}^{16\times32\times32}$ | Same spatial Gaussian posterior, represented by scalar fields |
| Learned parameters | 3,958,435 | 1,180,035 |
| Estimated convolutional work | $3.65 \times 10^{10}$ MAC/sample | $1.598 \times 10^{11}$ MAC/sample |
| Rotation constraint | None beyond translation-equivariant convolution | Continuous SO(2) kernel constraint |

### 3.8. VAE Objective and Training

Both models used a denoising VAE training criterion: an encoder sees a possibly corrupted patch, and the decoder is trained to reconstruct the corresponding clean patch (Im et al., 2017). In the implemented training procedure, with probability 0.3 the corruption operator $C$ perturbed hematoxylin/eosin/residual stain coordinates and added image-space Gaussian noise (Ruifrok and Johnston, 2001). The resulting input was clamped to the normalized $[-1, 1]$ RGB domain. A patch could therefore be clean or corrupted at the input, while its reconstruction target was always the same clean patch:

$$\mathbf{x} \xrightarrow{C} \widetilde{\mathbf{x}} \xrightarrow{q_{\boldsymbol{\phi}}} \mathbf{z} \xrightarrow{D_{\boldsymbol{\theta}}} \widehat{\mathbf{x}} \quad \text{compared with } \mathbf{x}. \tag{7}$$

The encoder distribution $q_{\boldsymbol{\phi}}(\mathbf{z} \mid \widetilde{\mathbf{x}})$ is the diagonal Gaussian determined by the mean and log-variance heads; $D_{\boldsymbol{\theta}}$ is the decoder. During training, $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}$, with $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The implemented loss was

$$\begin{aligned} \mathcal{L} = \; & \mathrm{MAE}(\widehat{\mathbf{x}}, \mathbf{x}) + 0.1[1 - \mathrm{SSIM}(\widehat{\mathbf{x}}, \mathbf{x})] \\ & + \beta D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z} \mid \widetilde{\mathbf{x}}) \,\|\, \mathcal{N}(\mathbf{0}, \mathbf{I})) \,, \end{aligned} \tag{8}$$

where MAE measures pixelwise reconstruction error, SSIM measures structural similarity (Wang et al., 2004), and the KL term regularizes the input-dependent posterior toward a unit Gaussian. The KL weight increased linearly from zero to 0.01 during the first 6,000 updates and remained fixed thereafter.

AdamW (Loshchilov and Hutter, 2019) used coefficients $(0.9, 0.999)$, $\epsilon = 10^{-8}$, weight decay $10^{-5}$ and gradient-norm clipping at 1.0. The learning rate increased linearly for 600

updates to $10^{-3}$ and then decreased with a cosine schedule to $10^{-5}$. Each model was trained for 60,000 successful updates with a global batch of 50 patches distributed over two NVIDIA T4 GPUs. Validation and checkpointing occurred every 3,000 updates. The comparison used the final update-60,000 checkpoint for both models rather than selecting different VAE checkpoints from the downstream results.

After VAE training, all encoder and decoder parameters were frozen. Downstream tasks used the posterior mean $\boldsymbol{\mu}$ rather than a sampled latent. For every selected coordinate, both encoders received the same RGB patch. The resulting paired latent datasets had identical WSI membership, tissue labels and patch coordinates.

### 3.9. Reconstruction Evaluation

The reconstruction test contained 67,138 foreground patches from the 23 test WSIs. For each slide, eligible foreground coordinates were sorted by row and column ($y$, then $x$). If there were more than 3,000, 3,000 evenly spaced rank positions were selected across that ordered list; otherwise all were retained. Thus, 22 slides contributed 3,000 patches each and one contributed its available 1,138. The cap limited the influence of slides with many more eligible patches. MAE and mean squared error (MSE) were measured in the normalized $[-1, 1]$ domain. Peak signal-to-noise ratio (PSNR) and SSIM were measured after conversion to $[0, 1]$ and clamping. Mean MAE was the primary reconstruction measure. The other metrics describe complementary error properties.

### 3.10. Classification A: Tissue Recognition and Label Budgets

Tissue classification uses one patch at a time, without combining information across a WSI. It tests how much tissue information each frozen representation retains as the number of labeled training patches changes. The task predicted tumor, stroma or necrosis from a single latent map. The classifier contained three convolutional stages $16 \to 32 \to 64 \to 128$, GroupNorm, GELU activations (Hendrycks and Gimpel, 2016), adaptive average pooling and a linear $128 \to 3$ output layer.

Training subsets contained 250, 500, 1,000, 2,500 and 5,671 labeled patches per class. They were class balanced, nested and selected without replacement; every smaller set was a subset of each larger one. WSI-stratified ordering ensured that the smallest subset did not collapse onto a single slide. Each normal/equivariant pair started from identical classifier weights and received the same patch order. Models were optimized for at most 30 epochs, and the best validation macro-F1 checkpoint was retained. The test distribution was not balanced and contained 21,796 tumor, 8,500 stroma and 1,276 necrosis patches from 23 WSIs.

### 3.11. Classification B: Five-Class WSI Diagnosis

The diagnostic task predicted the five ovarian-cancer subtypes. Every selected foreground patch was represented by its frozen posterior mean. A trainable convolutional patch encoder compressed the $16 \times 32 \times 32$ latent map. Two local attention blocks exchanged information within a radius-two spatial neighborhood, and a global stream summarized the complete WSI through learned sigmoid attention. A final classification layer produced five logits. The supervised model contained approximately 1.51 million parameters.

Table 3: Reconstruction on 67,138 test patches. Values are mean (population standard deviation).

| Metric | Conventional | SO(2) |
|---|---|---|
| MAE ↓ | 0.0629 (0.0239) | 0.0642 (0.0223) |
| MSE ↓ | 0.0080 (0.0074) | 0.0083 (0.0067) |
| PSNR dB ↑ | 27.923 (2.721) | 27.658 (2.595) |
| SSIM ↑ | 0.7575 (0.0842) | 0.7436 (0.0862) |

Training used one complete WSI bag per optimizer step, AdamW and class-weighted cross-entropy computed from the 106 training WSIs. Each representation branch started from the same classifier weights and received the same WSI order. The maximum was 150 epochs, validation was performed twice per epoch, and each branch independently retained its highest validation macro-F1 checkpoint. The primary test metric was macro-F1; balanced accuracy, accuracy, mean cross-entropy, classwise F1 and confusion matrices were secondary summaries.

### 3.12. Statistical Analysis

All difference intervals are reported as conventional-minus-equivariant. Reconstruction differences used 10,000 paired cluster-bootstrap replicates at WSI level (Field and Welsh, 2007). WSI diagnosis used 10,000 paired bootstrap replicates stratified by diagnosis. Tissue recognition used a paired WSI bootstrap stratified by observed tissue support. The five label-budget macro-F1 differences and the area under the log-budget learning curve (AULC) difference were considered together as six related comparisons. Their 95% intervals were adjusted simultaneously: across the six comparisons, the bootstrap procedure targets 95% coverage for the whole set of differences. A difference interval entirely below zero favors the equivariant model when higher is better; an interval entirely above zero favors the conventional model. An interval containing zero does not establish a statistically significant difference at the 5% level, nor does it establish equivalence. Patch standard deviations describe variation among patches, while confidence intervals resample WSIs as the biological units.

## 4. Results

### 4.1. VAE Reconstruction

Figure 8 displays the first nine patches from the fixed set of 25 validation patches. Each original is paired with both frozen models' final reconstruction.

The conventional VAE had lower pooled MAE and MSE and higher pooled PSNR and SSIM (Table 3). Mean MAE was 0.062872 for the conventional model and 0.064230 for the equivariant model. The paired difference was $-0.001358$, with a 95% interval of $[-0.002213, 0.000046]$. The interval included zero, so the primary analysis did not establish a reconstruction difference. The remaining metrics are descriptive secondary summaries.

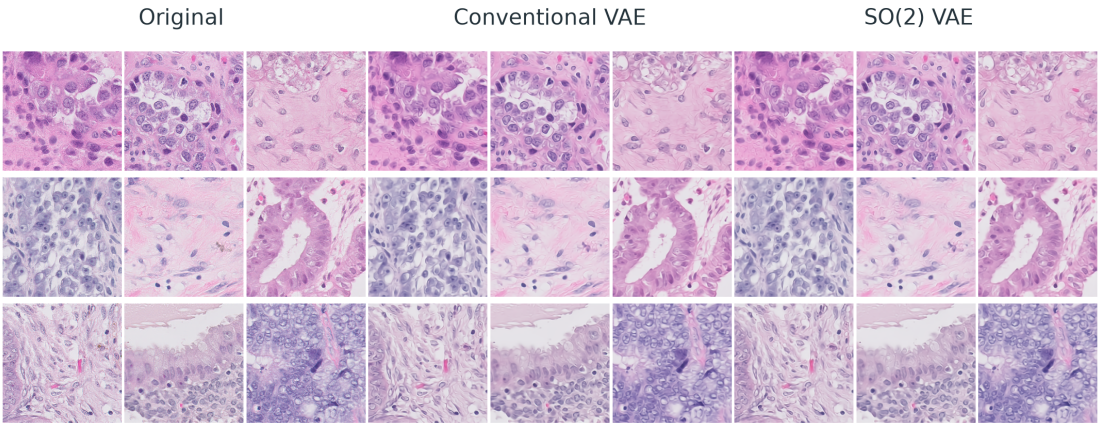



Figure 8: Three matched $3 \times 3$ mosaics: original patches, conventional-VAE reconstructions and continuous-SO(2)-VAE reconstructions. The nine examples are the first nine from the fixed set of 25 validation patches; quantitative results use 67,138 patches from 23 test WSIs.

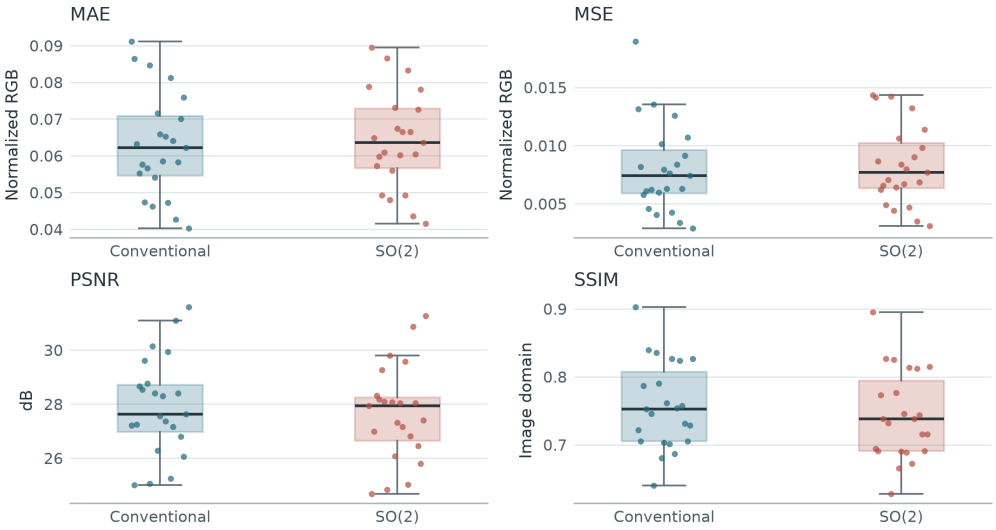



Figure 9: Reconstruction metric distributions across 23 test WSIs. Each point is one WSI-level mean; boxes summarize those 23 means for each frozen VAE. They are descriptive distributions, not confidence intervals. MAE and MSE use normalized RGB; PSNR and SSIM use image-domain values.

Figure 9 shows descriptive variation among the 23 test WSIs. Each plotted value is the mean of that WSI's selected test patches; the paired WSI-bootstrap interval above remains the inferential comparison.

Table 4: Tissue macro-F1 on 31,572 test patches. Simultaneous intervals are reported for conventional-minus-SO(2).

| Labels/class | Conventional ↑ | SO(2) ↑ | Simultaneous 95% CI |
|---|---|---|---|
| 250 | 0.508 | 0.523 | $[-0.072, 0.041]$ |
| 500 | 0.511 | 0.575 | $[-0.121, -0.008]$ |
| 1,000 | 0.582 | 0.598 | $[-0.072, 0.041]$ |
| 2,500 | 0.701 | 0.714 | $[-0.069, 0.044]$ |
| 5,671 | 0.766 | 0.710 | $[-0.001, 0.112]$ |

### 4.2. Classification A: Three-Class Tissue Recognition

The tissue results changed with the amount of labeled training data (Figure 10 and Table 4). The equivariant representation had higher macro-F1 at 250, 500, 1,000 and 2,500 labels per class. At 500 labels per class, the macro-F1 values were 0.575 and 0.511, and the simultaneous interval for conventional-minus-equivariant was $[-0.121, -0.008]$. This was the only label budget whose simultaneous interval excluded zero.

At 500 labels per class, the classwise F1 values for conventional/equivariant representations were 0.769/0.781 for tumor, 0.485/0.560 for stroma and 0.279/0.384 for necrosis. These values show how the aggregate macro-F1 difference was distributed across the three tissue classes.

At the largest budget, the conventional representation obtained macro-F1 0.766 compared with 0.710 for the equivariant representation. The simultaneous interval included zero, with a lower bound of $-0.001$. The corresponding classwise F1 values were 0.868/0.871 for tumor, 0.728/0.711 for stroma and 0.701/0.548 for necrosis. The normalized area under the log-budget learning curve was 0.615 for the conventional representation and 0.631 for the equivariant representation; its simultaneous difference interval included zero.

### 4.3. Classification B: Five-Class WSI Diagnosis

Macro-F1 was 0.483 for the equivariant representation and 0.389 for the conventional representation (Table 5). The conventional-minus-equivariant difference was $-0.095$, with a 95% interval of $[-0.207, 0.050]$. The interval included zero. Among the secondary metrics, the equivariant representation had lower mean cross-entropy, with a WSI-resampling interval excluding zero.

Classwise point estimates varied (Table 6). The conventional representation had zero F1 for EC and LGSC and correctly classified both MC slides; the equivariant representation had higher EC and LGSC F1 and no correct MC prediction. With only two test WSIs in each of LGSC and MC, these contrasts are unstable.

## 5. Discussion

Which representation scored higher depended on the task and the number of labeled tissue patches. The conventional VAE had lower pooled MAE/MSE and higher pooled

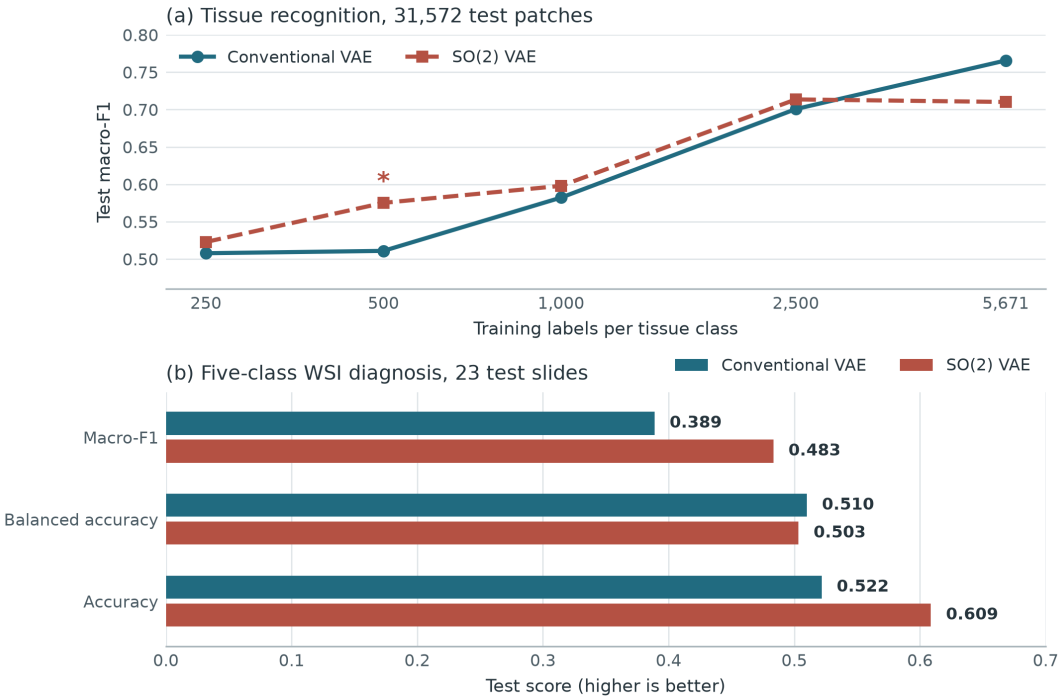



Figure 10: Frozen-representation classification results. (a) Tissue macro-F1 on 31,572 test patches under nested class-balanced training budgets; the asterisk marks the 500-label contrast whose simultaneous 95% difference interval excludes zero. (b) Five-class diagnosis on 23 test WSIs. The paired WSI macro-F1 interval includes zero.

Table 5: Overall WSI diagnosis results on 23 test slides. Difference intervals are conventional-minus-SO(2).

| Metric | Conventional | SO(2) | Difference 95% CI |
|---|---|---|---|
| Macro-F1 ↑ | 0.389 | 0.483 | $[-0.207, 0.050]$ |
| Balanced accuracy ↑ | 0.510 | 0.503 | $[-0.167, 0.180]$ |
| Accuracy ↑ | 0.522 | 0.609 | $[-0.217, 0.043]$ |
| Mean cross-entropy ↓ | 1.323 | 0.993 | $[0.135, 0.524]$ |

PSNR/SSIM point estimates. The equivariant representation had higher tissue macro-F1 at the four smaller budgets and a higher WSI macro-F1 point estimate. Most confidence intervals for the main comparisons included zero. The single tissue-budget interval that excluded zero applies to that contrast only and does not establish a monotone or general label-efficiency effect. The model with lower reconstruction error did not consistently achieve higher classification macro-F1.

Table 6: Classwise WSI F1 and test support.

| Diagnosis | Support | Conventional F1 ↑ | SO(2) F1 ↑ |
|---|---|---|---|
| CC | 5 | 0.667 | 0.750 |
| EC | 6 | 0.000 | 0.533 |
| HGSC | 8 | 0.706 | 0.800 |
| LGSC | 2 | 0.000 | 0.333 |
| MC | 2 | 0.571 | 0.000 |

Patch-based WSI systems depend on both the instance representation and the bag aggregator. Prior MIL work has developed the representation, aggregation or both (Ilse et al., 2018; Lu et al., 2021; Li et al., 2021; Shao et al., 2021). This experiment held the downstream protocol fixed while changing the frozen representation. The WSI interval remained wide, and classwise estimates were unstable for diagnoses with two test slides. The results characterize the two implemented representation branches but do not establish a general WSI or tissue-classification advantage.

The models also differ in capacity and computation. Graham *et al.* matched parameter counts in comparisons of rotation-equivariant histology classifiers (Graham et al., 2020); these two VAEs were not parameter-matched. The equivariant VAE uses 29.8% as many learned parameters but requires more estimated multiply-accumulate operations. Any performance difference may reflect equivariance, parameter count, optimization or their interaction. The experiment does not separate these mechanisms.

## 6. Limitations

Each VAE and downstream configuration was trained once. The bootstrap intervals quantify variation across the observed WSIs, not variation across new initializations, data orders or optimization runs. The WSI test contains 23 slides, including two each for LGSC and MC. Tissue recognition contains many patches but only 23 WSIs, and necrosis is present in five of them.

Supplemental tissue masks are incomplete. The tissue experiment evaluates selected high-purity annotated regions and cannot be interpreted as exhaustive segmentation of complete WSIs. The study uses one ovarian-cancer data source.

## 7. Conclusion

This study compared conventional and continuous-SO(2) VAEs as frozen histopathology representations under a common experimental protocol. With one training run per configuration and intervals limited to WSI resampling, the results varied by task and label budget. The experiment did not establish a general advantage for either implemented representation; repeated training runs and models with matched parameter counts are needed to isolate the contribution of continuous rotation equivariance.

## References

James Burgess, Jeffrey J. Nirschl, Maria-Clara Zanellati, Alejandro Lozano, Sarah Cohen, and Serena Yeung-Levy. Orientation-invariant autoencoders learn robust representations for shape profiling of cells and organelles. *Nature Communications*, 15:1022, 2024. doi: 10.1038/s41467-024-45362-4.

Taco S. Cohen and Max Welling. Group equivariant convolutional networks. In *Proceedings of the 33rd International Conference on Machine Learning*, pages 2990–2999, 2016.

Taco S. Cohen and Max Welling. Steerable CNNs. In *International Conference on Learning Representations*, 2017.

Bryn Elesedy and Sheheryar Zaidi. Provably strict generalisation benefit for equivariant models. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 2959–2969, 2021.

Stefan Elfwing, Eiji Uchibe, and Kenji Doya. Sigmoid-weighted linear units for neural network function approximation in reinforcement learning. *Neural Networks*, 107:3–11, 2018. doi: 10.1016/j.neunet.2017.12.012.

Christopher A. Field and Alan H. Welsh. Bootstrapping clustered data. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 69(3):369–390, 2007. doi: 10.1111/j.1467-9868.2007.00593.x.

Simon Graham, David Epstein, and Nasir Rajpoot. Dense steerable filter CNNs for exploiting rotational symmetry in histology images. *IEEE Transactions on Medical Imaging*, 39 (12):4124–4136, 2020. doi: 10.1109/TMI.2020.3013246.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 770–778, 2016. doi: 10.1109/CVPR.2016.90.

Dan Hendrycks and Kevin Gimpel. Gaussian error linear units (GELUs), 2016.

Maximilian Ilse, Jakub M. Tomczak, and Max Welling. Attention-based deep multiple instance learning. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80, pages 2127–2136, 2018.

Daniel Jiwoong Im, Sungjin Ahn, Roland Memisevic, and Yoshua Bengio. Denoising criterion for variational auto-encoding framework. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 31, pages 2059–2065, 2017. doi: 10.1609/aaai.v31i1.10777.

Diederik P. Kingma and Max Welling. Auto-encoding variational bayes. In *International Conference on Learning Representations*, 2014.

Maxime W. Lafarge, Josien P. W. Pluim, and Mitko Veta. Orientation-disentangled unsupervised representation learning for computational pathology. *arXiv preprint arXiv:2008.11673*, 2020. URL https://arxiv.org/abs/2008.11673.

Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998. URL https://leon.bottou.org/papers/lecun-98h.

Bin Li, Yin Li, and Kevin W. Eliceiri. Dual-stream multiple instance learning network for whole slide image classification with self-supervised contrastive learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14318–14328, 2021.

Geert Litjens, Thijs Kooi, Babak Ehteshami Bejnordi, Arnaud Arindra Adiyoso Setio, Francesco Ciompi, Mohsen Ghafoorian, Jeroen A. W. M. van der Laak, Bram van Ginneken, and Clara I. Sanchez. A survey on deep learning in medical image analysis. *Medical Image Analysis*, 42:60–88, 2017. doi: 10.1016/j.media.2017.07.005.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019.

Ming Y. Lu, Drew F. K. Williamson, Tiffany Y. Chen, Richard J. Chen, Matteo Barbieri, and Faisal Mahmood. Data-efficient and weakly supervised computational pathology on whole-slide images. *Nature Biomedical Engineering*, 5:555–570, 2021. doi: 10.1038/s41551-020-00682-w.

Alireza Nasiri and Tristan Bepler. Unsupervised object representation learning using translation and rotation group equivariant VAE. In *Advances in Neural Information Processing Systems*, volume 35, 2022. doi: 10.52202/068431-1110.

Nobuyuki Otsu. A threshold selection method from gray-level histograms. *IEEE Transactions on Systems, Man, and Cybernetics*, 9(1):62–66, 1979. doi: 10.1109/TSMC.1979.4310076.

Arnout C. Ruifrok and Dennis A. Johnston. Quantification of histochemical staining by color deconvolution. *Analytical and Quantitative Cytology and Histology*, 23(4):291–299, 2001.

Zhuchen Shao, Hao Bian, Yang Chen, Yifeng Wang, Jian Zhang, Xiangyang Ji, and Yongbing Zhang. TransMIL: Transformer based correlated multiple instance learning for whole slide image classification. In *Advances in Neural Information Processing Systems*, volume 34, pages 2136–2147, 2021.

sohier. UBC Ovarian Cancer Competition Supplemental Masks. Kaggle dataset, 2023. URL https://www.kaggle.com/datasets/sohier/ubc-ovarian-cancer-competition-supplemental-masks.

Andrew H. Song, Guillaume Jaume, Drew F. K. Williamson, Ming Y. Lu, Anurag Vaidya, Tiffany R. Miller, and Faisal Mahmood. Artificial intelligence for digital and computational pathology. *Nature Reviews Bioengineering*, 1(12):930–949, 2023. doi: 10.1038/s44222-023-00096-8.

David Tellez, Geert Litjens, Jeroen van der Laak, and Francesco Ciompi. Neural image compression for gigapixel histopathology image analysis. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(2):567–578, 2021. doi: 10.1109/TPAMI.2019.2936841.

University of British Columbia. UBC ovarian cancer subtype classification and outlier detection (UBC-OCEAN). Kaggle competition, 2023. URL https://www.kaggle.com/competitions/UBC-OCEAN.

Bastiaan S. Veeling, Jasper Linmans, Jim Winkens, Taco Cohen, and Max Welling. Rotation equivariant CNNs for digital pathology. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2018*, pages 210–218. Springer, 2018. doi: 10.1007/978-3-030-00934-2_24.

Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004. doi: 10.1109/TIP.2003.819861.

Maurice Weiler and Gabriele Cesa. General $E(2)$-equivariant steerable CNNs. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

Daniel E. Worrall, Stephan J. Garbin, Daniyar Turmukhambetov, and Gabriel J. Brostow. Harmonic networks: Deep translation and rotation equivariance. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017. doi: 10.1109/CVPR.2017.758.